

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin
437171483	1.4735	4.947%	923	73.37	3"	Clean
411457490	2.4324	5.531%	431	58.00	1"	Alias
295373509	5.4219	8.026%	247	22.12	2"	Clean
374158907	14.6563	4.974%	179	21.40	2"	Alias

TIC 111240209

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 33.245106 d (ALIASED, see below)
-> corrected period: 66.490212 d
Epoch (t0) 2240.2131 BTJD
Duration 0.2440 d
Depth 16.214 % (162,137 ppm)
SNR 1156.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

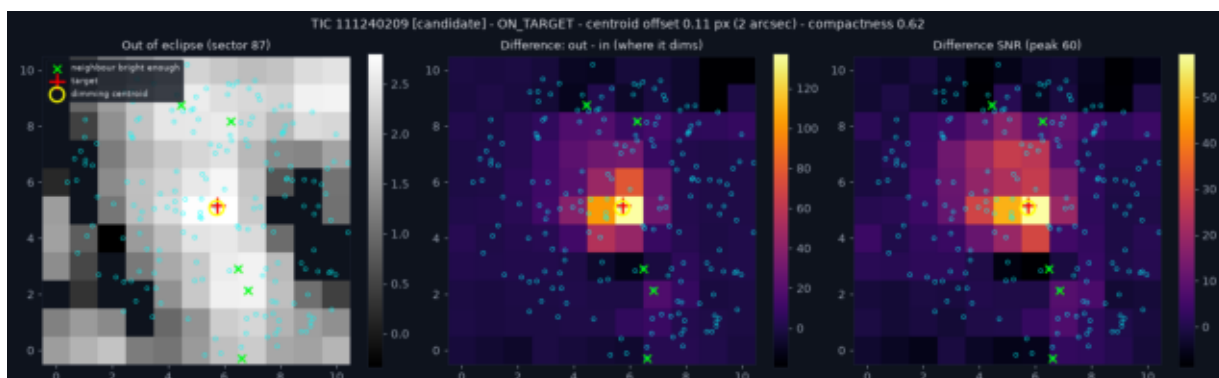
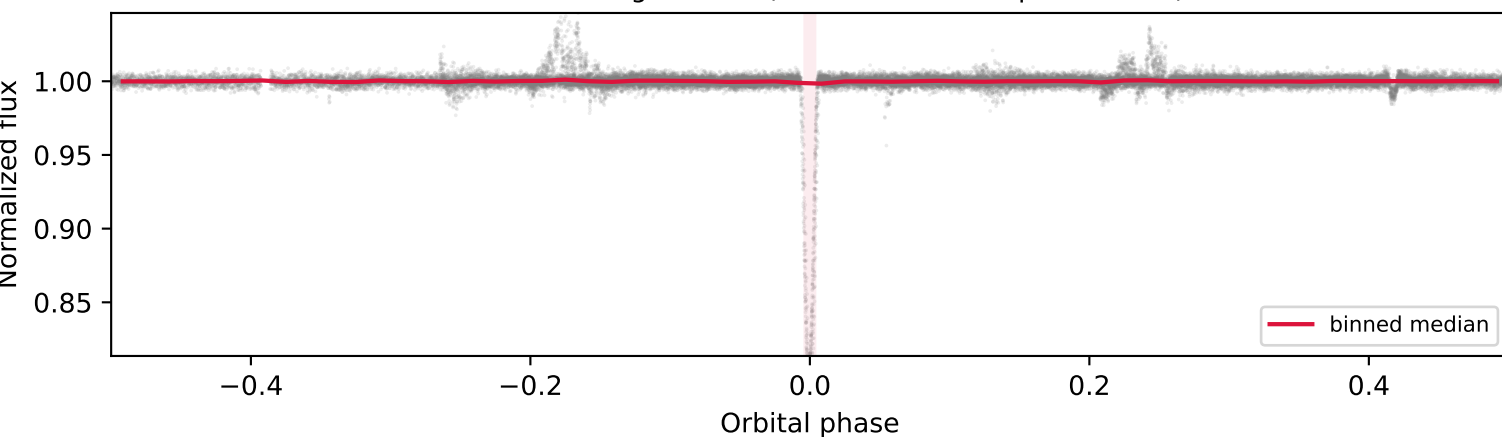
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 903.5 (gate ≥ 7.0)
Odd/even z-score 39.49 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 87 light-curve dip in this sector: 101.3 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 59.53 (gate ≥ 5.0) Compactness 0.62 (gate ≥ 0.50)
Dimming centroid offset 0.11 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 111240236
2.29 px away, dTmag +1.10 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 39.49 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 66.490212 d, in any write-up.

TIC 239661046

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.053991 d
Epoch (t0) 1816.9715 BTJD
Duration 0.0580 d
Depth 6.249 % (62,488 ppm)
SNR 889.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

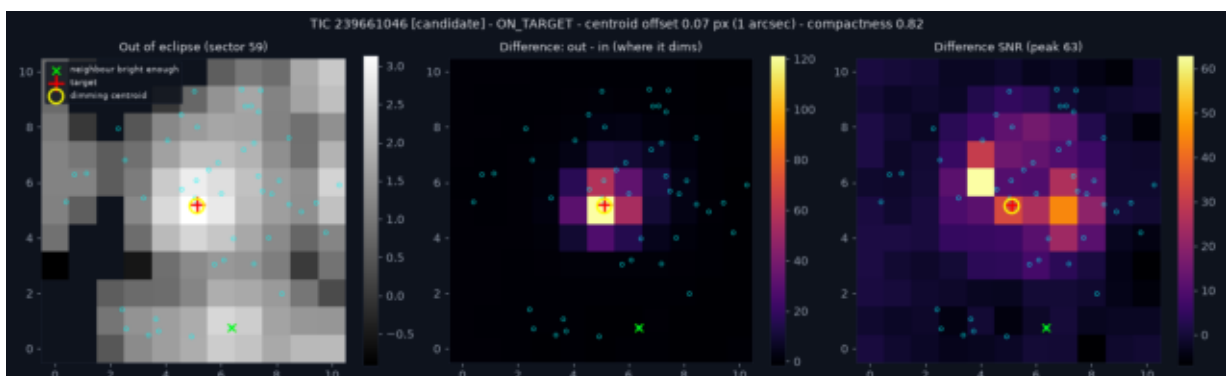
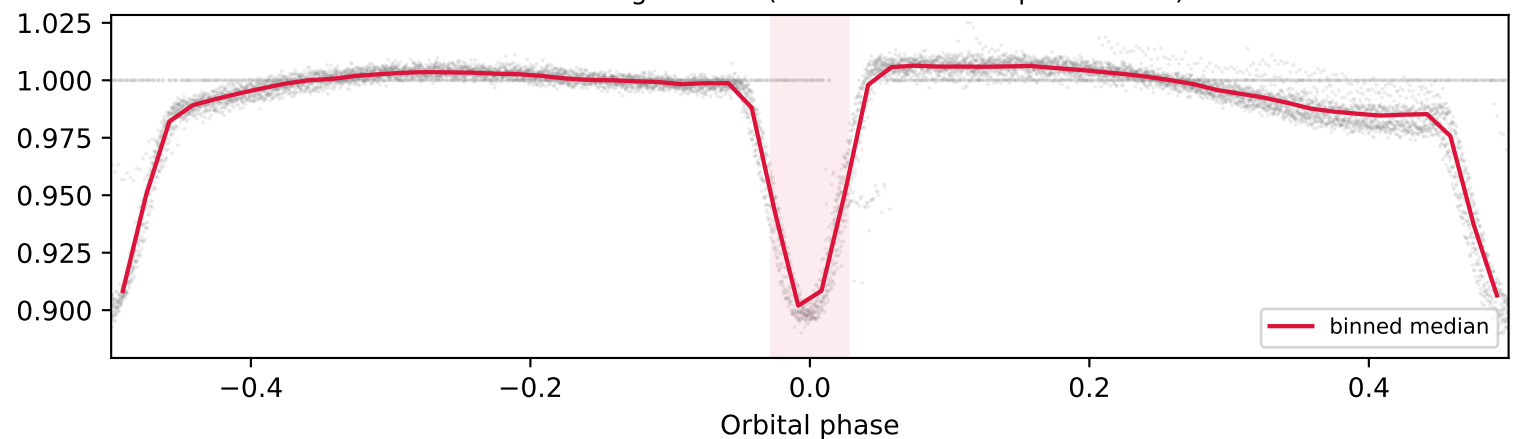
Eclipses observed 36 (gate ≥ 3)
Robust depth ratio 1.32 (gate ≥ 0.50)
Robust depth sigma 248.6 (gate ≥ 7.0)
Odd/even z-score 0.50 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 59 light-curve dip in this sector: 409.3 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 63.02 (gate ≥ 5.0) Compactness 0.82 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 239661034
4.57 px away, dTmag +1.39 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



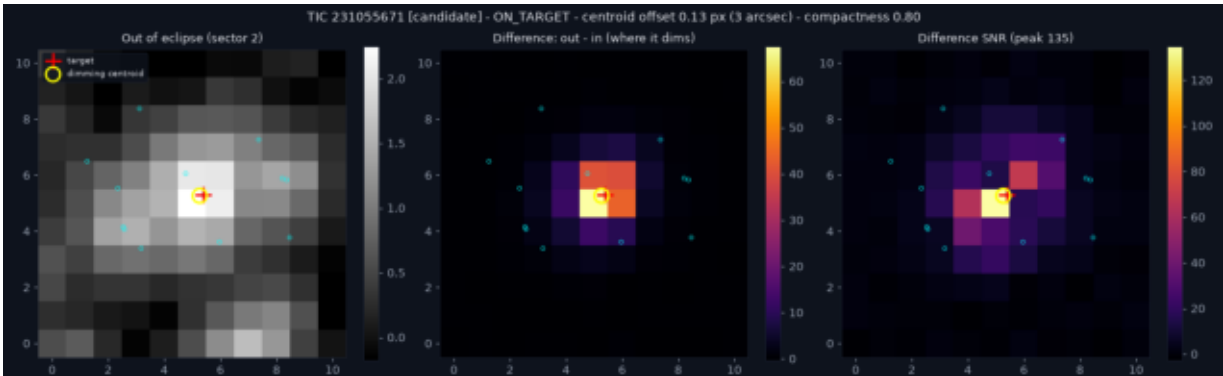
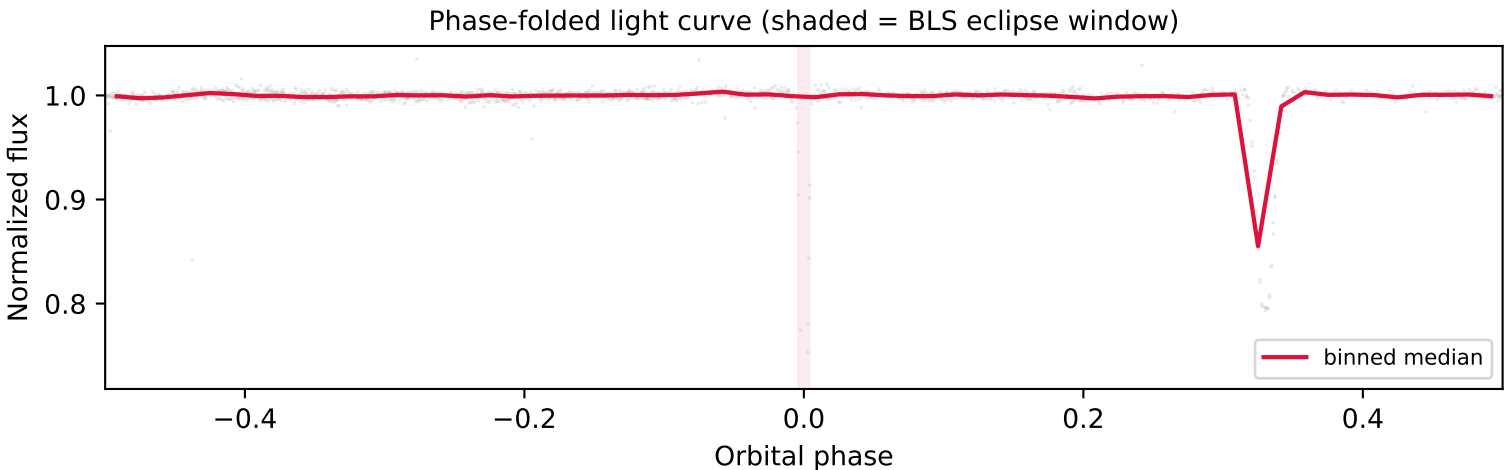
No caveats - clean detection on every check.

TIC 231055671

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	15.261462 d (ALIASED, see below)	VSX	no match
-> corrected period:	30.522924 d	ASAS-SN	no match
Epoch (t0)	1361.8447 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	32.573 % (325,727 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	652.2	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	3 (gate >= 3)		
Robust depth ratio	1.03 (gate >= 0.50)		
Robust depth sigma	302.9 (gate >= 7.0)		
Odd/even z-score	6.42 (ALIAS FLAGGED)		
Secondary eclipse	detected, phase 0.34, 7.5 sigma		
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)			
Sector 2	light-curve dip in this sector: 265.2 sigma	t0 refined by 0.0000 phase	
Difference-image peak SNR	134.81 (gate >= 5.0)	Compactness	0.80 (gate >= 0.50)
Dimming centroid offset	0.13 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)		
Nearest bright-enough neighbour in cutout:	none within the cutout		



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.42 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 30.522924 d, in any write-up.

TIC 273787849

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 2.142549 d
Epoch (t0) 1327.4683 BTJD
Duration 0.0840 d
Depth 2.803 % (28,030 ppm)
SNR 402.7

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 54 (gate ≥ 3)
Robust depth ratio 1.01 (gate ≥ 0.50)
Robust depth sigma 154.4 (gate ≥ 7.0)
Odd/even z-score 2.06 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

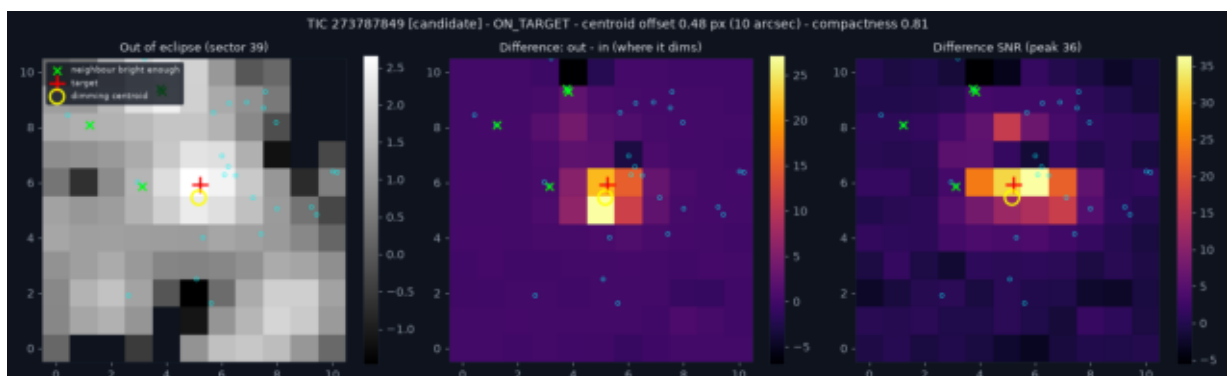
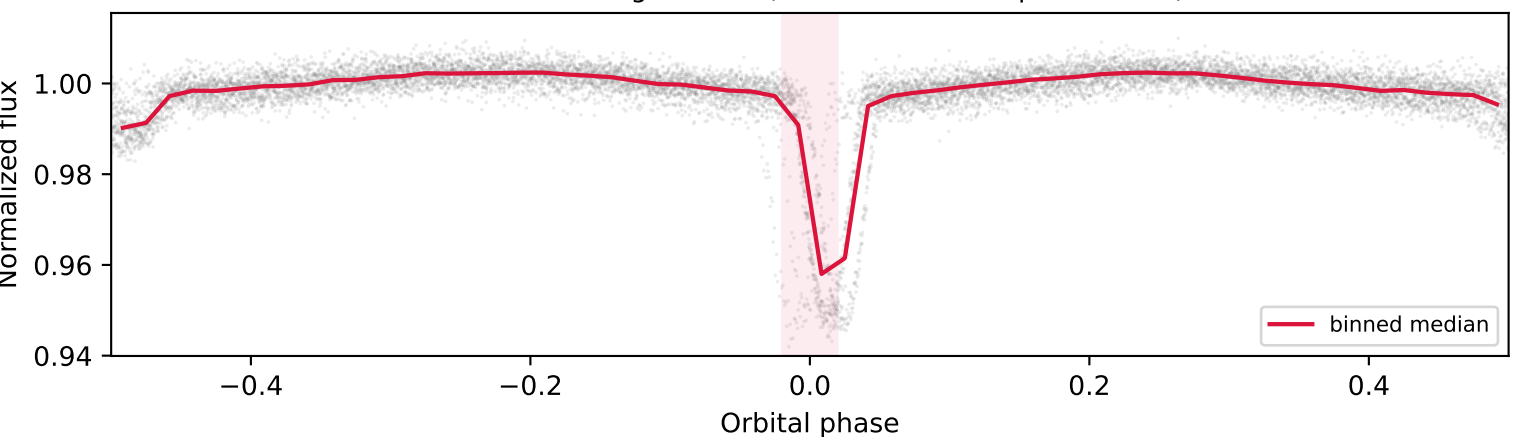
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 39 light-curve dip in this sector: 151.4 sigma t0 refined by 0.0250 phase
Difference-image peak SNR 36.41 (gate ≥ 5.0) Compactness 0.81 (gate ≥ 0.50)
Dimming centroid offset 0.48 px (10 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 273787858
2.08 px away, dTmag +3.47 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

TIC 453692345

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.814639 d
Epoch (t0) 1570.6181 BTJD
Duration 0.0580 d
Depth 9.708 % (97,077 ppm)
SNR 397.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

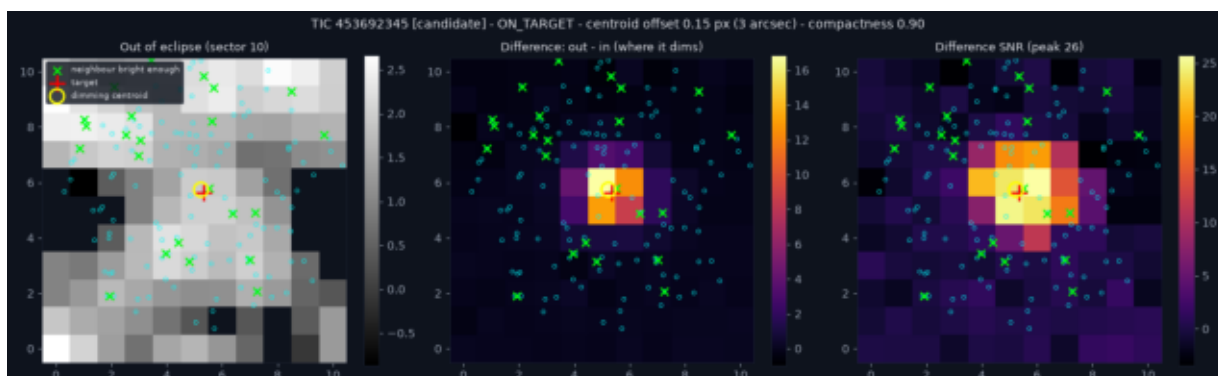
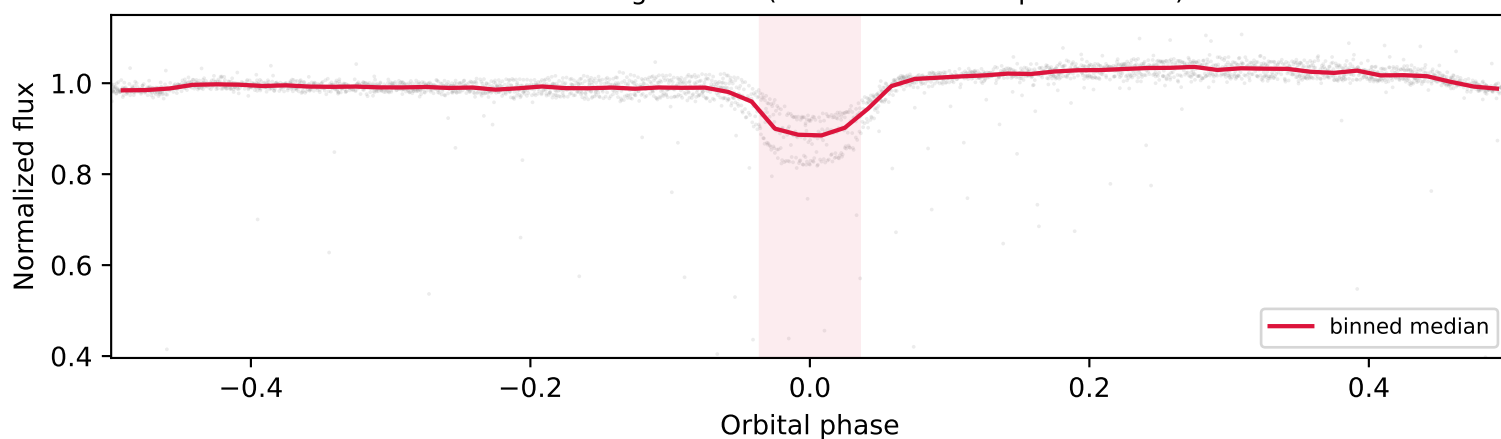
Eclipses observed 59 (gate ≥ 3)
Robust depth ratio 1.10 (gate ≥ 0.50)
Robust depth sigma 58.8 (gate ≥ 7.0)
Odd/even z-score 1.85 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 25.9 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 25.75 (gate ≥ 5.0) Compactness 0.90 (gate ≥ 0.50)
Dimming centroid offset 0.15 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 453692346
0.29 px away, dTmag +1.69 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



TIC 150230753

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	2.334152 d (ALIASED, see below)	VSX	no match
-> corrected period:	4.668305 d	ASAS-SN	no match
Epoch (t0)	1626.0382 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	16.455 % (164,549 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	235.7	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	7 (gate >= 3)		
Robust depth ratio	1.01 (gate >= 0.50)		
Robust depth sigma	53.0 (gate >= 7.0)		
Odd/even z-score	4.17 (ALIAS FLAGGED)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

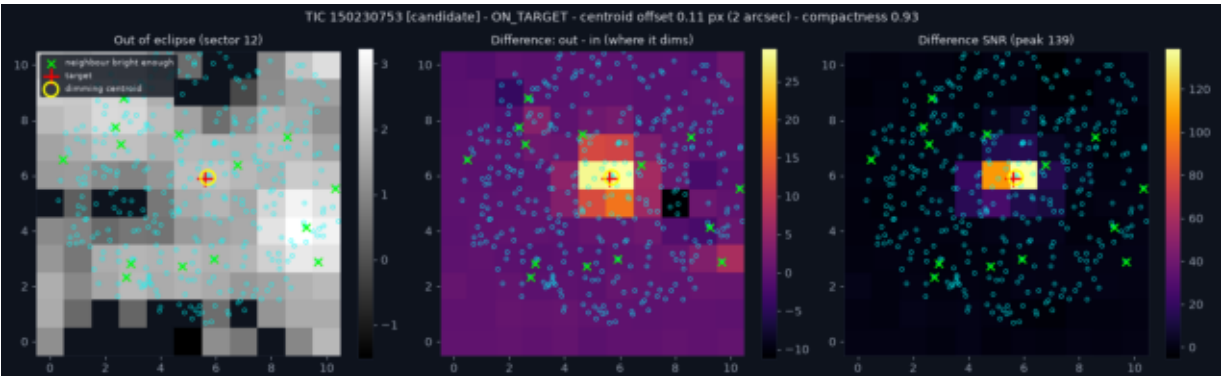
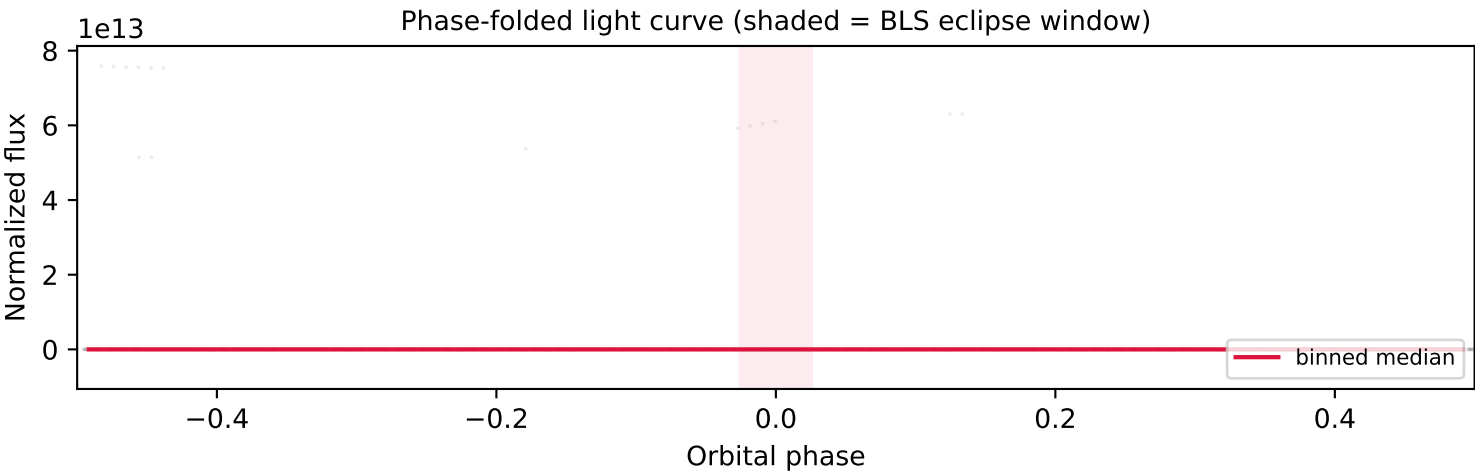
Sector 12 light-curve dip in this sector: 50.2 sigma t0 refined by -0.0075 phase

Difference-image peak SNR 139.03 (gate >= 5.0) Compactness 0.93 (gate >= 0.50)

Dimming centroid offset 0.11 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 150237925

1.16 px away, dTmag +0.72 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.668305 d, in any write-up.

TIC 158398880

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 3.430730 d (ALIASSED, see below)
-> corrected period: 6.861461 d
Epoch (t0) 1656.8732 BTJD
Duration 0.2440 d
Depth 5.166 % (51,657 ppm)
SNR 174.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

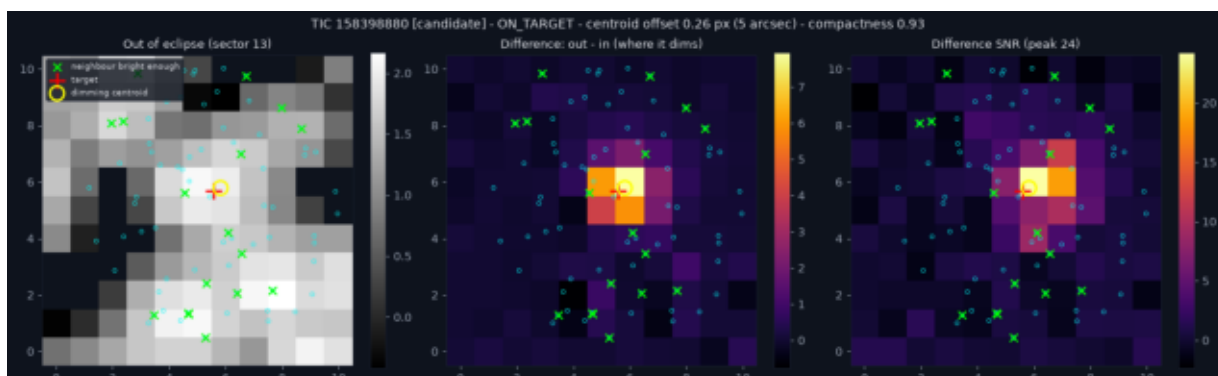
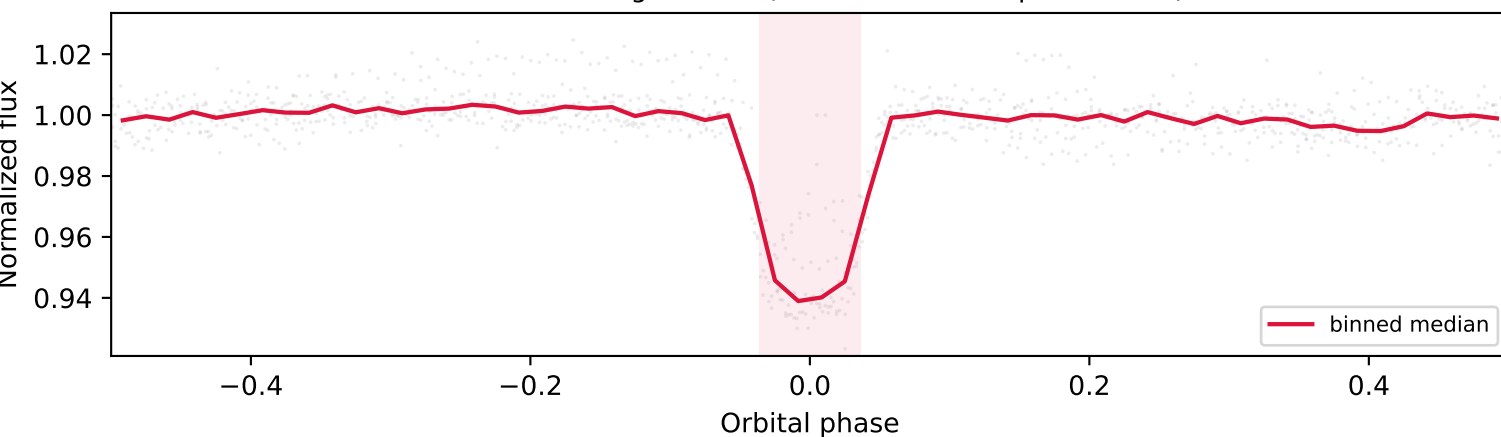
Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 1.10 (gate ≥ 0.50)
Robust depth sigma 88.0 (gate ≥ 7.0)
Odd/even z-score 6.23 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 95.0 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 24.28 (gate ≥ 5.0) Compactness 0.93 (gate ≥ 0.50)
Dimming centroid offset 0.26 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 158398881
1.28 px away, dTmag +0.05 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 6.23 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 6.861461 d, in any write-up.